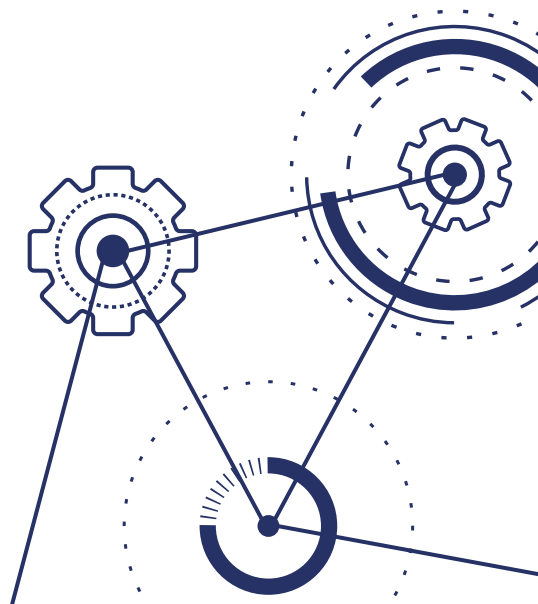




Design Review

Ola, Elysia, Lindsay, Evan, Miles, Nathan

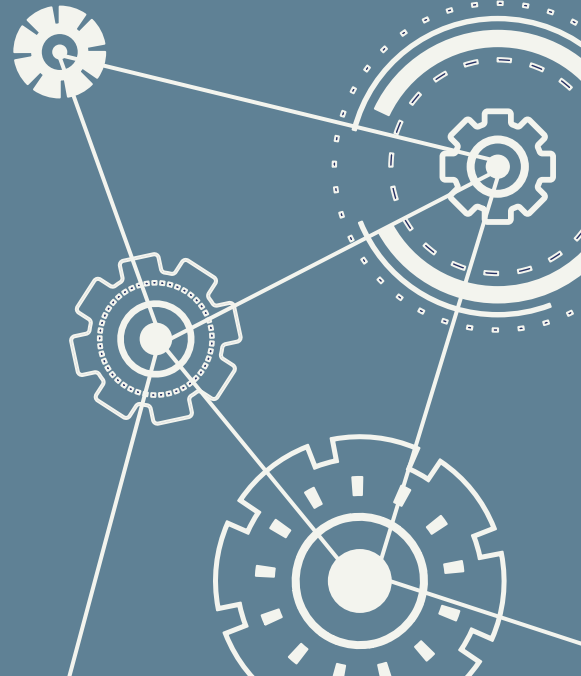


Criteria

Presentation Structure - delete slide after done

• please include clear depictions of a cross section of the body/ring assembly, highlighting the press fit details, through the use of screenshots from CAD
• include an exploded view of the yo-yo assembly with all designed parts (missing thermoform)
• outline how many parts and molds your team intends to make
• include a summary of each of your CAM programs (in progress)
• students will be graded on ○ completeness of the required elements ○ clarity of presentation

Design Overview

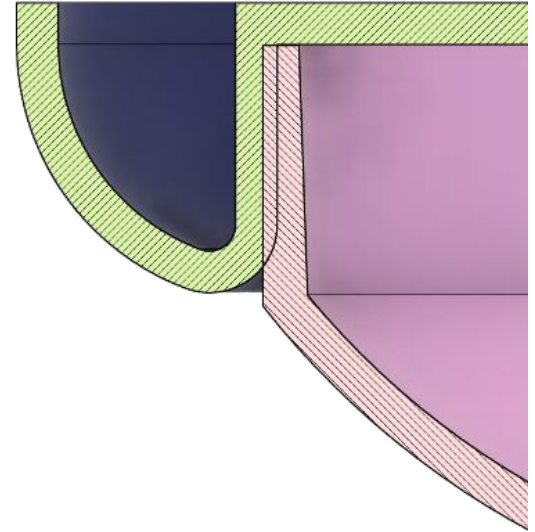
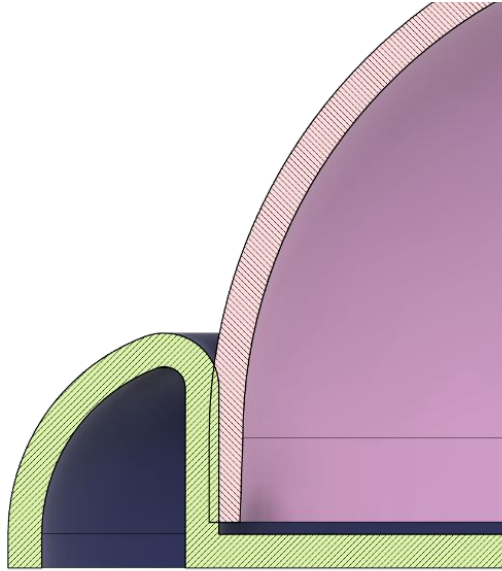
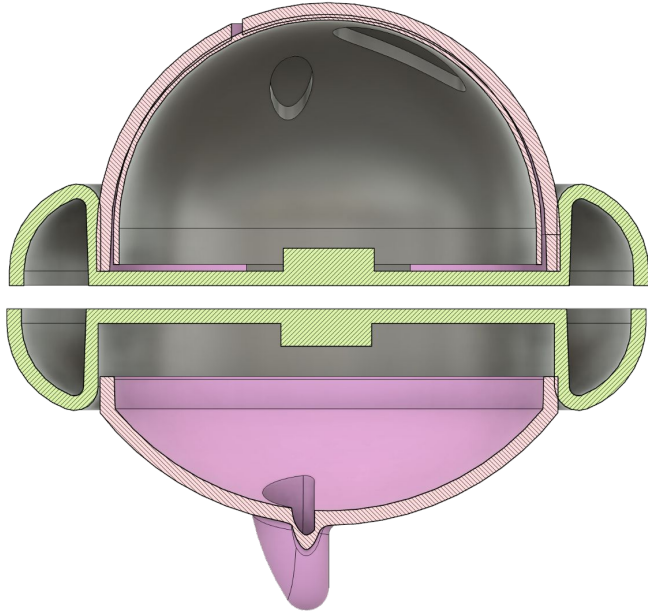
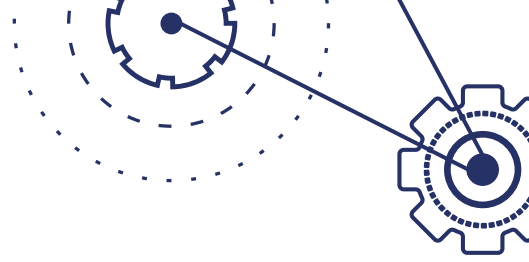


Inspiration



CAD - press fit

cross section of body/ring assembly

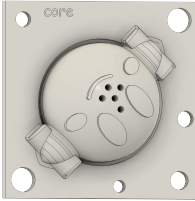
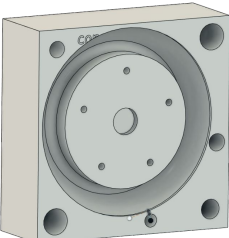
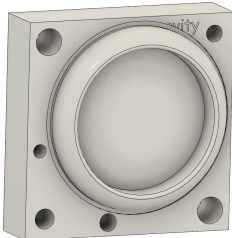


CAD - exploded view



Kir-B

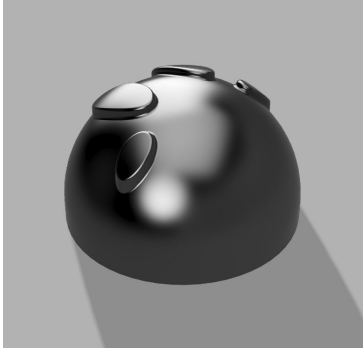
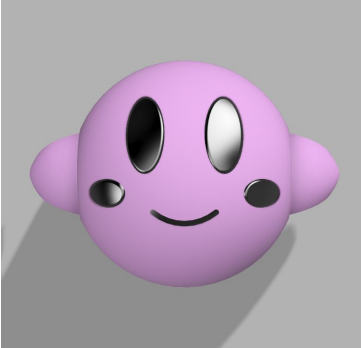
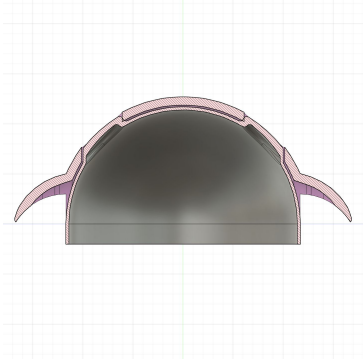
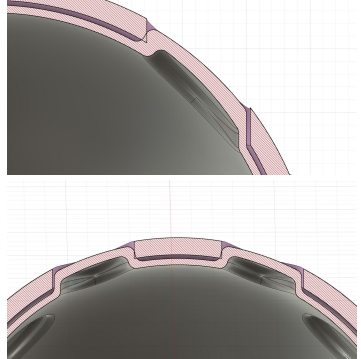
Yo-yo IM part outline

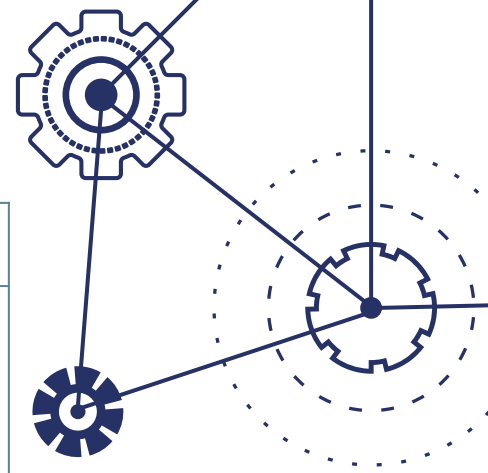
KirB Part (quantity)	Core	Cavity
Upper Body (1)	 A 3D model of the Upper Body Core, showing a square block with a central circular cavity and four small holes on the top face. The word "core" is visible on the top face.	 A 3D model of the Upper Body Cavity, showing a square block with a central circular cavity and four small holes on the top face. The word "cavity" is visible on the top face.
Lower Body (1)	 A 3D model of the Lower Body Core, showing a square block with a central circular cavity and four small holes on the top face. The word "core" is visible on the top face.	 A 3D model of the Lower Body Cavity, showing a square block with a central circular cavity and four small holes on the top face. The word "cavity" is visible on the top face.
Floatie [yo-yo body] (2)	 A 3D model of the Floatie Core, showing a square block with a central circular cavity and four small holes on the top face. The word "core" is visible on the top face.	 A 3D model of the Floatie Cavity, showing a square block with a central circular cavity and four small holes on the top face. The word "cavity" is visible on the top face.

Manufacturing Molds and Thermo Form



Yo-yo TF part outline

KirB Part (quantity)	SLA Printed Mold
Face (1)	   



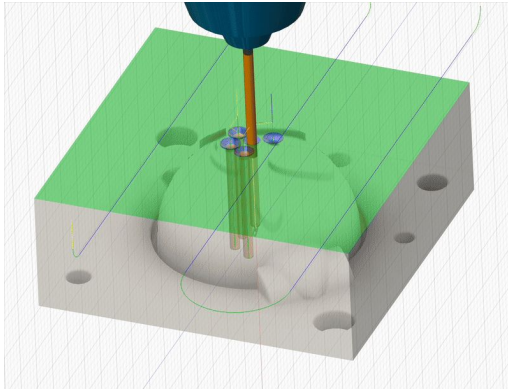
Kir-B

CAM Programs

Upper Body - core

Time: 2:57 hrs

Operation	Instance	Setup	Work Offset	Tool
 Face2	1/1	Setup3	2	#14 - Ø2" 45° face
 Drill2	1/1	Setup3	2	#11 - Ø1/4" 130° spot drill (SPOT DRILL)
 Drill3	1/1	Setup3	2	#12 - Ø0.12" drill (Ejector Pin Drill)
 Adaptive2	1/1	Setup3	2	#5 - Ø1/4" flat (End Mill)
 Contour3	1/1	Setup3	2	#5 - Ø1/4" flat (End Mill)
 Morphed ...	1/1	Setup3	2	#3 - Ø1/8" flat (End Mill)
 2D Contour2	1/1	Setup3	2	#10 - Ø1/16" 2° tapered (2 Degree Tapered End Mill)
 Flat1	1/1	Setup3	2	#5 - Ø1/4" flat (End Mill)



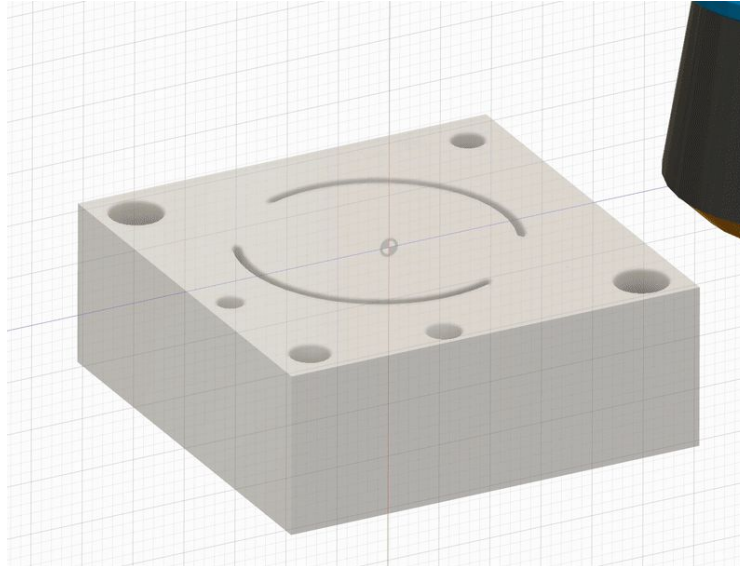
Kir-B

CAM Programs

Upper Body - cavity

Time: 1:35 hrs

Operation	Instance	Setup	Work Offset	Tool
 Face1	1/1	cavity	3	#14 - Ø2" 45° face
 Adaptive1	1/1	cavity	3	#6 - Ø3/8" flat (End Mill)
 Contour1	1/1	cavity	3	#8 - Ø1/8" ball (End Mill)
 Contour2	1/1	cavity	3	#5 - Ø1/4" flat (End Mill)
 2D Contour1	1/1	cavity	3	#8 - Ø1/8" ball (End Mill)

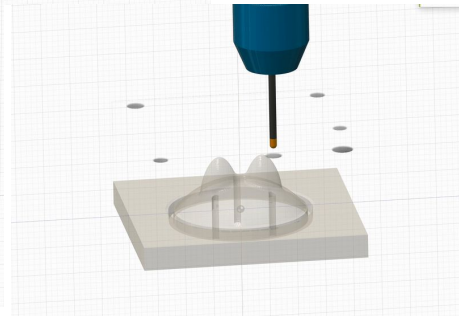
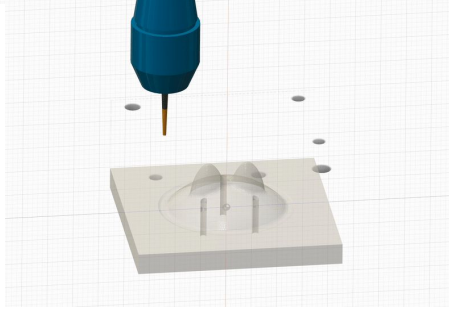
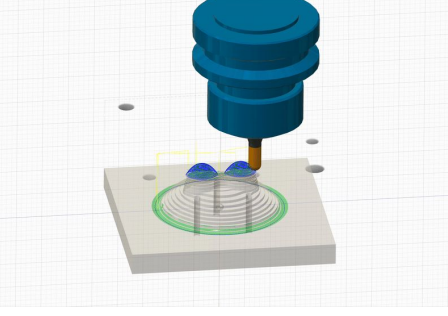
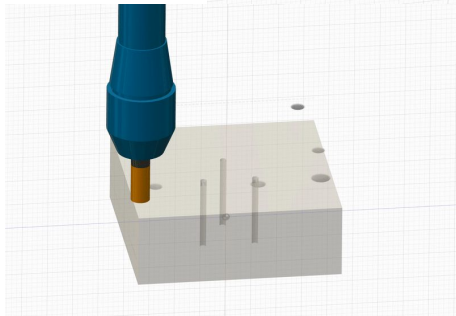
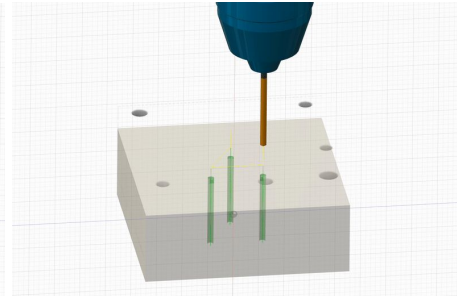
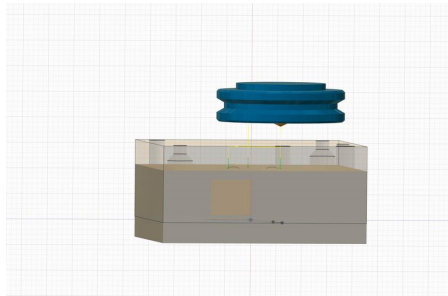
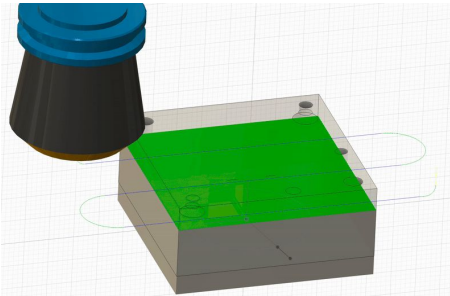


CAM Programs

Lower Body - core

Time: 1:02 hrs

Operation	Instance	Setup	Work Offset	Tool
 Face1	1/1	core	2	#14 - Ø2" 45° face
 Drill1	1/1	core	2	#11 - Ø1/4" 130° spot drill (SPOT DRILL)
 Drill2	1/1	core	2	#12 - Ø0.12" drill (Ejector Pin Drill)
 Adaptive2	1/1	core	2	#6 - Ø3/8" flat (End Mill)
 Contour2	1/1	core	2	#9 - Ø1/4" ball (End Mill)
 2D Pocket1	1/1	core	2	#10 - Ø1/16" 2° tapered (2 Degree Tapered End Mill)
 2D Contour1	1/1	core	2	#8 - Ø1/8" ball (End Mill)

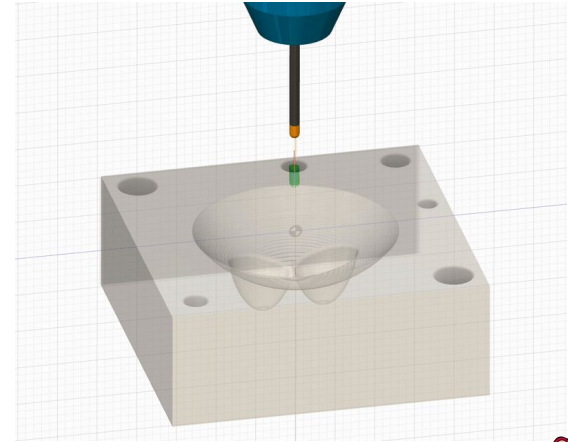
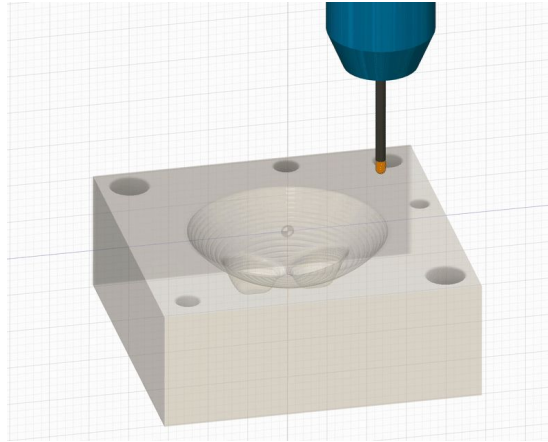
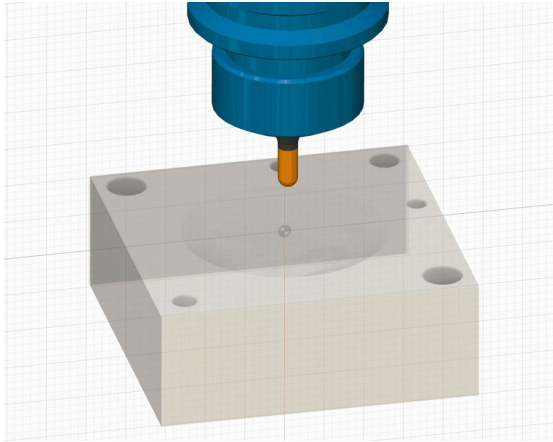


CAM Programs

Lower Body - cavity

Time: 51 mins

Operation	Instance	Setup	Work Offset	Tool
 Face2	1/1	cavity	3	#14 - Ø2" 45° face
 Adaptive4	1/1	cavity	3	#9 - Ø1/4" ball (End Mill)
 Contour3	1/1	cavity	3	#8 - Ø1/8" ball (End Mill)
 2D Contour2	1/1	cavity	3	#8 - Ø1/8" ball (End Mill)



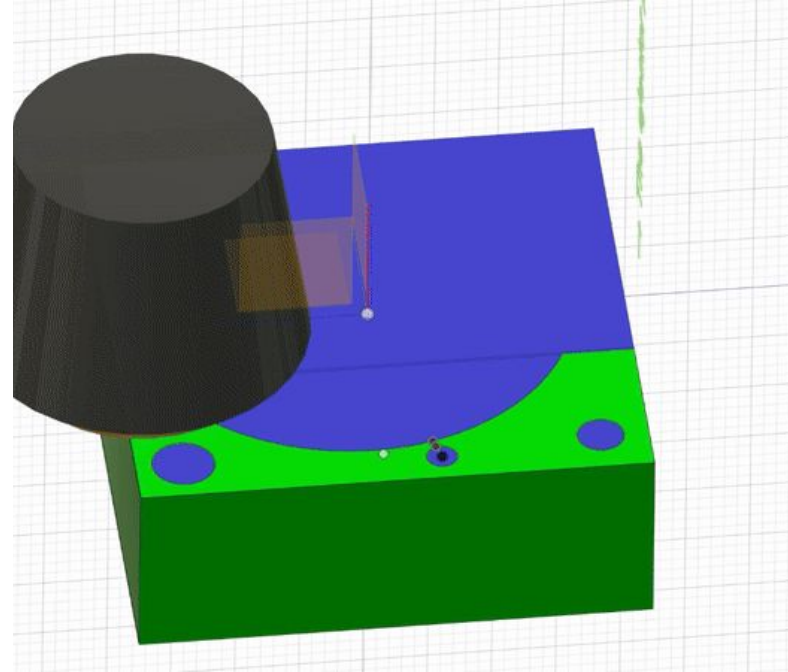
Kir-B

CAM Programs

Floatie - core

Time: 1:08 hrs

Operation	Instance	Setup	Work Offset	Tool
 Face5	1/1	Setup7	2	#14 - Ø2" 45° face
 Bore1	1/1	Setup7	2	#6 - Ø3/8" flat (End Mill)
 Drill1	1/1	Setup7	2	#11 - Ø1/4" 130° spot drill (SPOT DRILL)
 Drill3	1/1	Setup7	2	#12 - Ø0.12" drill (Ejector Pin Drill)
 Adaptive7	1/1	Setup7	2	#5 - Ø1/4" flat (End Mill)
 Adaptive2	1/1	Setup7	2	#4 - Ø3/16" flat (End Mill)
 Contour4	1/1	Setup7	2	#9 - Ø1/4" ball (End Mill)
 Contour6	1/1	Setup7	2	#8 - Ø1/8" ball (End Mill)
 Flat1	1/1	Setup7	2	#6 - Ø3/8" flat (End Mill)
 2D Contour4	1/1	Setup7	2	#8 - Ø1/8" ball (End Mill)

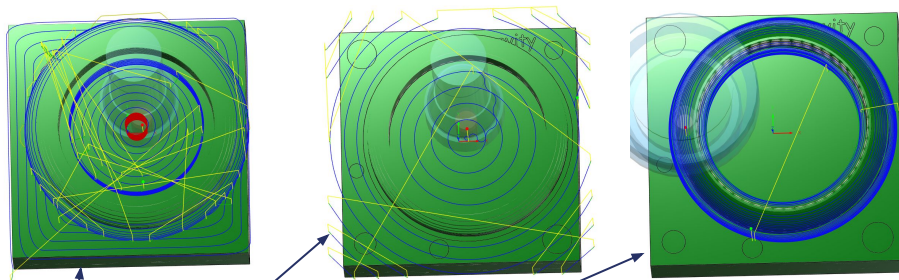


Kir-B

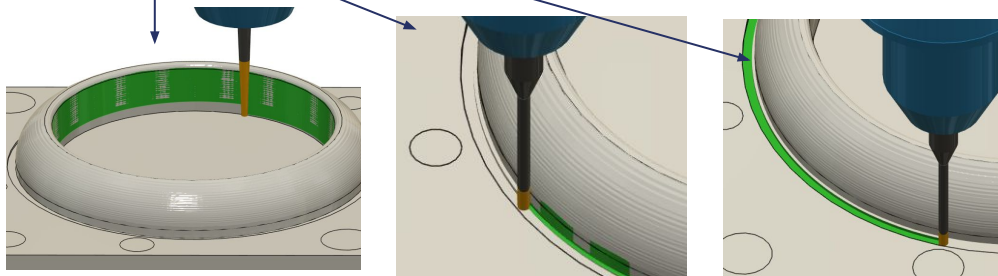
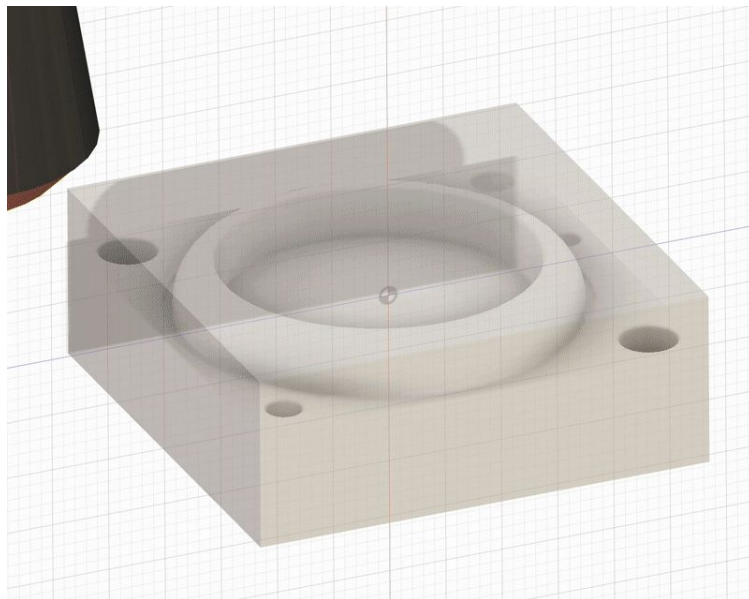
CAM Programs

Floatie - cavity

Time: 55 mins



Operation	Instance	Setup	Work Offset	Tool
Face1	1/1	cavity	3	#14 - Ø2" 45° face
Adaptive4	1/1	cavity	3	#6 - Ø3/8" flat (End Mill)
Flat1	1/1	cavity	3	#6 - Ø3/8" flat (End Mill)
Contour1	1/1	cavity	3	#9 - Ø1/4" ball (End Mill)
Bore1	1/1	cavity	3	#1 - Ø1/16" flat (End Mill)
Bore2	1/1	cavity	3	#1 - Ø1/16" flat (End Mill)
Bore3	1/1	cavity	3	#10 - Ø1/16" 2° tapered (2 Degree Tapered End Mill)



Kir-B

Criteria

Design for Manufacturing - delete slide after done

• are the non-press-fit features of your yo-yo properly designed for manufacturing? a. have you made features that are difficult or impossible to manufacture with high quality?
• are there draft angles in your design?
• wall thickness reasonably uniform?
• do all molds have all of the required features a. sprue, ejector pins, gates, runners
• are the CAM programs likely to lead to successful outcomes and have reasonable machine times? Review this when talking about the individual CAMs
• instructors will review the viability of insert-molding nuts with your design
• students will be graded on: ○ effectiveness and appropriateness of designs and programs

Design for Manufacturing

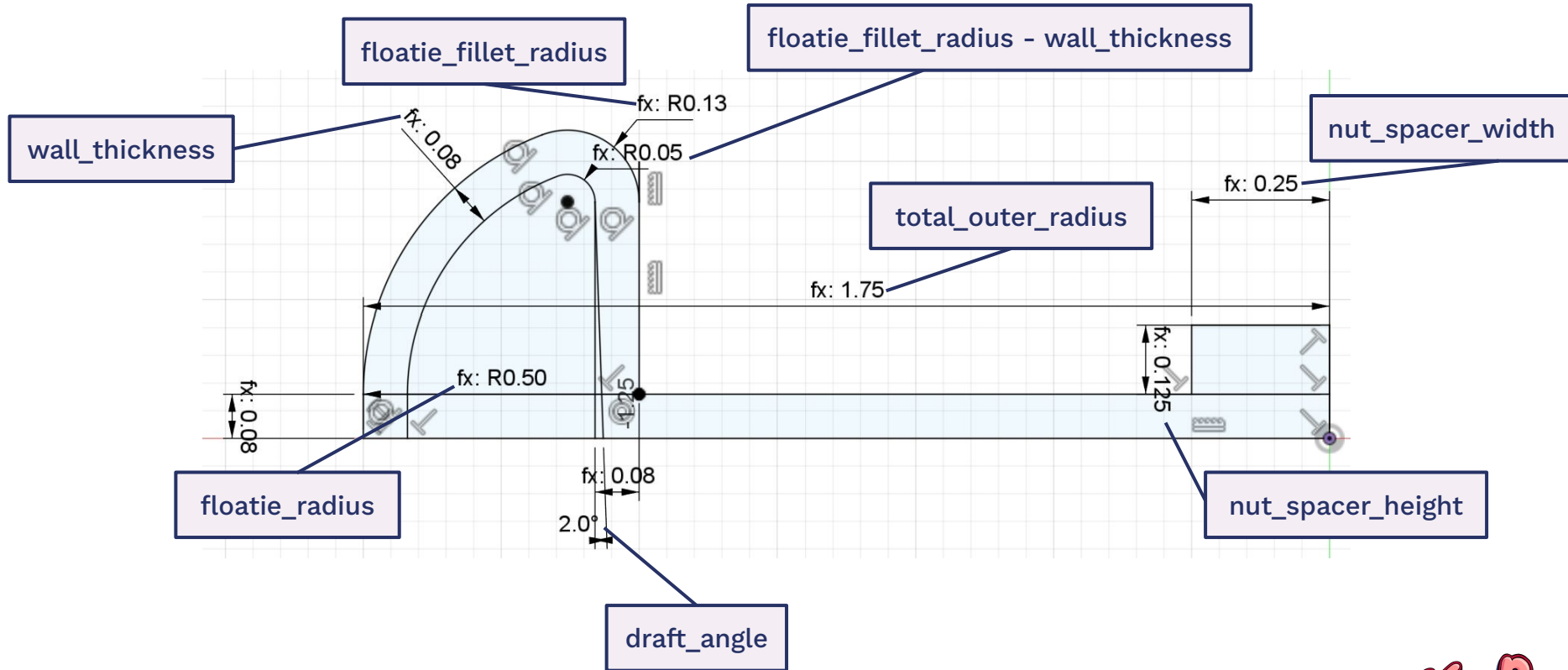


Our Parameters

Name	Unit	Expression	Value	Comments
wall_thickness	in	0.08 in	0.08	uniform wall thickness
outer_radius	in	1.25 in	1.25	outer radius of kirby
inner_radius	in	outer_radius - wall_thickness	1.17	inner radius of kirby
length_of_engagement	in	0.2 in	0.2	press fit engagement
total_outer_radius	in	outer_radius + floatie_radius	1.75	total size of floatie
nut_spacer_width	in	0.25 in	0.25	space for nut spacer
nut_spacer_height	in	(1 / 8) * 1 in	0.125	space for the nut spacer
floatie_radius	in	0.50 in	0.5	radius of just the floatie
interference	in	0.0242 in	0.024	interference for press fit
foot_heel_radius	in	0.45 in	0.45	radius of the circle composing kirby's foot
foot_length	in	1.65 in	1.65	length of kirby's foot from heel to toe
shrinkage_scaling		(1.0152 in) / in	1.0152	PP shrinkage
draft_angle	deg	2 deg	2	draft angle
floatie_fillte_radius	in	0.13 in	0.13	top edge fillet

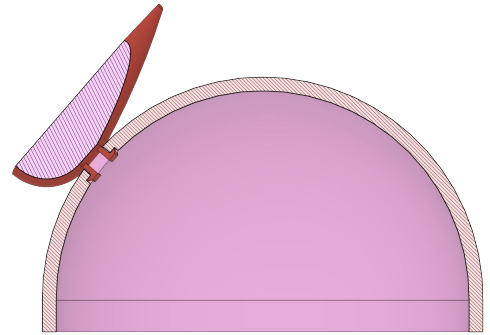
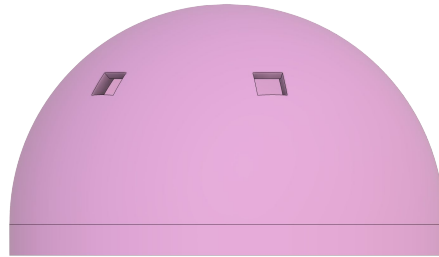
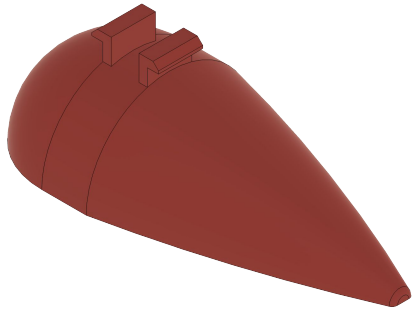
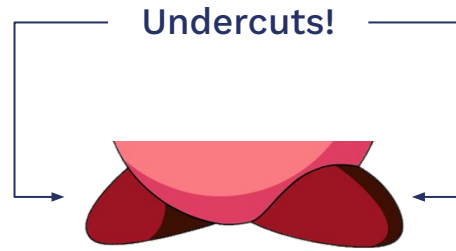


Parameters in use



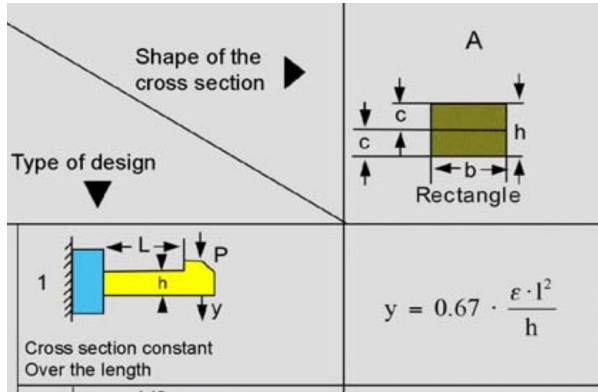
DFM

Lower Body

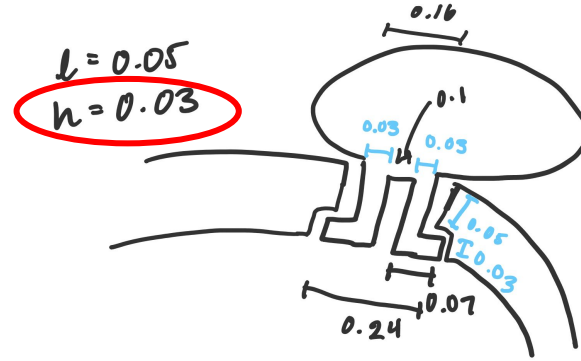


DFM

Lower Body



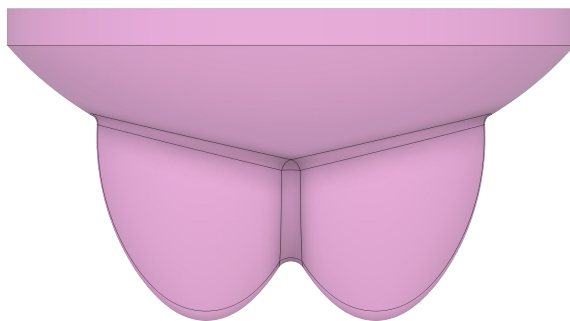
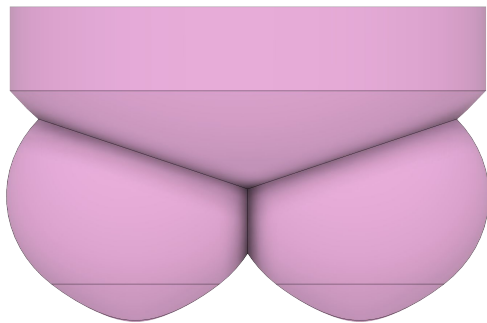
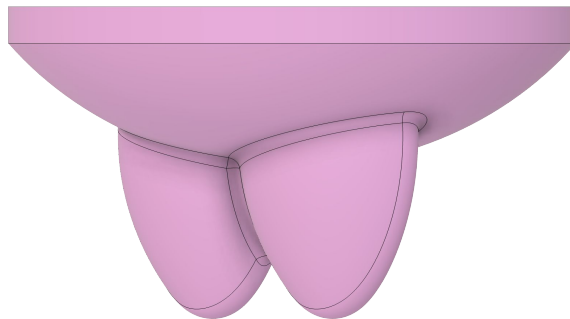
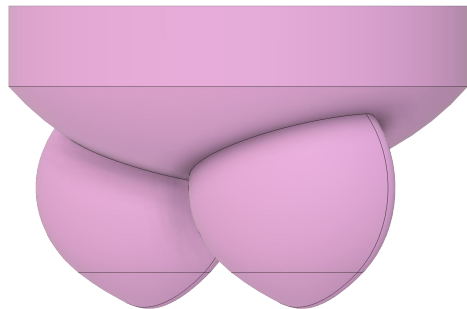
Snap-Fit Joints for Plastics - A Design Guide, Bayer Materials Science



⚙	Name ^	Corner radius	Diameter
▼	2.008 TOOL LIBRARY - S25		
	1 - Ø1/16" (End Mill)	0.0625 in	
	2 - Ø3/32" (End Mill)		
	3 - Ø1/8" (End Mill)		
	4 - Ø3/16" (End Mill)		
	5 - Ø1/4" (End Mill)		

DFM

Lower Body



V1 - Undercuts Galore

V2 - Extra Long

V3 - Short n' Sweet

One Body: Reduce Complexity + Improve Aesthetics



DFM

Upper Body



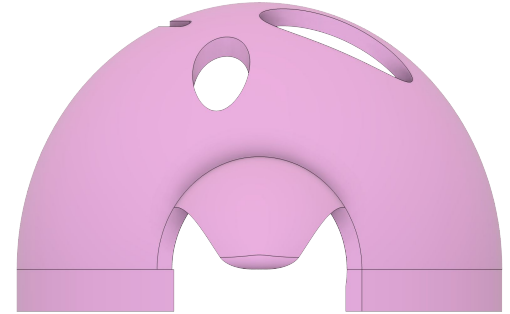
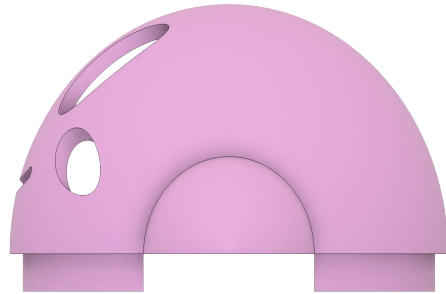
V1 - Reachable Material?



V2 - Awkward Floatie Interface



V3 - Ready to Go

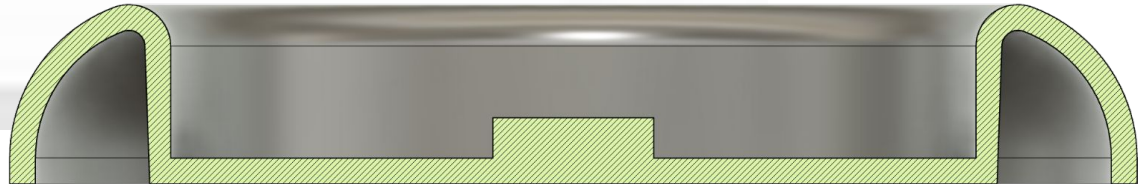
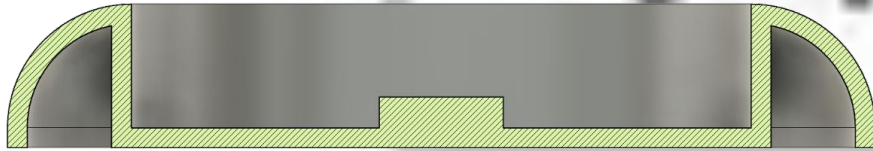
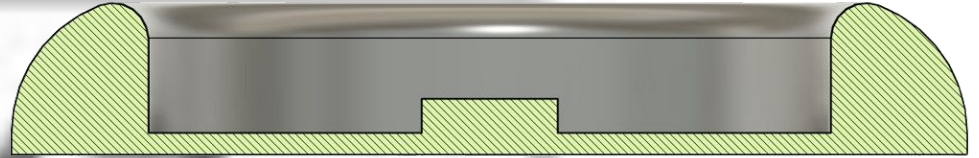


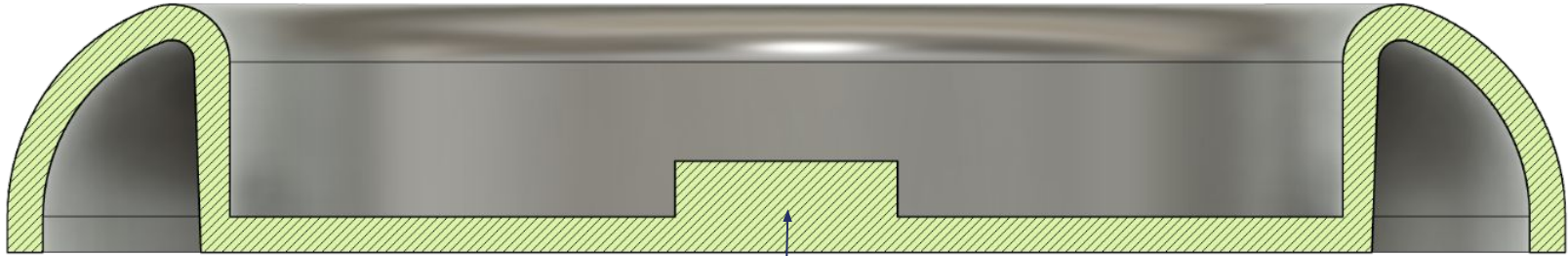
One Body: Reduce Complexity + Improve Aesthetics



DFM

floatie





Draft angles!
2 degrees

Largest nominal
variation in wall
thickness is 0.125 in

Criteria

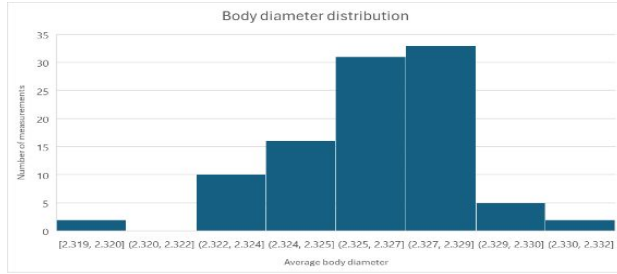
Press Fit Engineering Analysis - delete slide after done

- | |
|---|
| <ul style="list-style-type: none">• present the analysis documented in the Manufacturing Spreadsheet, preparing slides that highlight each key decision making step (all tabs highlighted in purple)<ul style="list-style-type: none">a. instructors and LAs will have the spreadsheet open for reference |
| <ul style="list-style-type: none">• show how all of your parts in CAD have been properly scaled to match your shrinkage modeling |
| <ul style="list-style-type: none">• students will be graded on:<ul style="list-style-type: none">a. the completeness of the spreadsheet parameters up to this pointb. the soundness of the engineering reasoning used to make key decisions about press fit parametersc. consistency between the manufacturing spreadsheet, CAD, and the message conveyed in the presentation |

Press Fit & Shrinkage Analysis

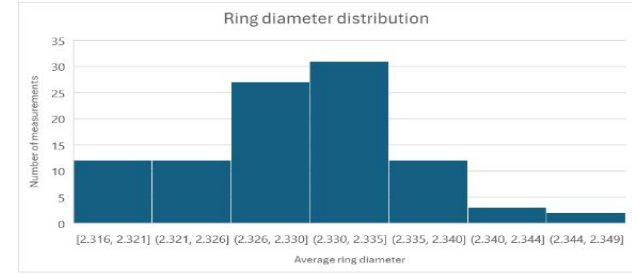


Press Fit Analysis

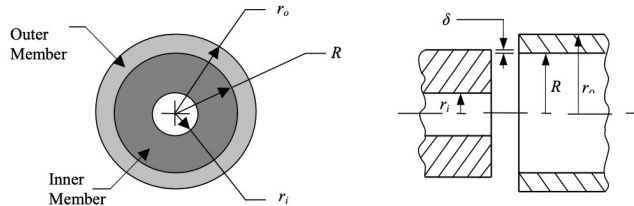


Empirical

Interference	
Team A	0.005131666
Team B	-0.0035
Team C	0.024495667
Team D	0.007320333
Team E	0.016



2.008 Yo-yo Projects - Spring 2024 Production Data



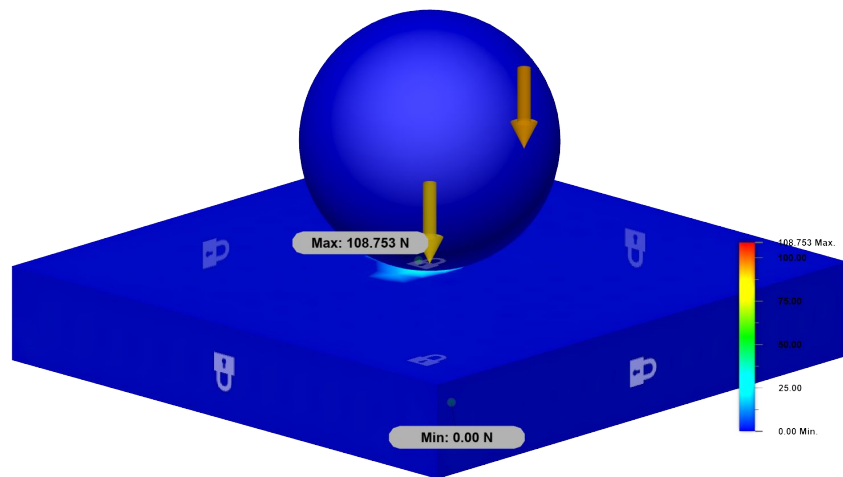
Theoretical

$$\delta_{\text{total}} = [K_o + K_i] p R$$

$$K_i = \frac{1}{E_i} \left[\left(\frac{R^2 + r_i^2}{R^2 - r_i^2} \right) - \nu_i \right] = \frac{1}{E_i} [C_i - \nu_i] \quad C_i = \frac{\xi_i^2 + 1}{\xi_i^2 - 1} \quad \xi_i = R / r_i$$

$$K_o = \frac{1}{E_o} \left[\left(\frac{r_o^2 + R^2}{r_o^2 - R^2} \right) + \nu_o \right] = \frac{1}{E_o} [C_o + \nu_o] \quad C_o = \frac{\xi_o^2 + 1}{\xi_o^2 - 1} \quad \xi_o = r_o / R$$

Press Fit Analysis



Typical Engineering Properties of Polypropylene

General Properties	English Units	SI Units
CAS Number	9003-07-0	9003-07-0
Density		
Homopolymer	56.4 - 56.6 Lbs /ft ³	0.904 - 0.908 g/cm ³
Random Copolymer	56.4 - 56.6 Lbs /ft ³	0.904 - 0.908 g/cm ³
Impact Copolymer	56.0 - 56.2 Lbs /ft ³	0.898 - 0.900 g/cm ³
TPOs	54.6 - 56.1 Lbs /ft ³	0.875 - 0.880 g/cm ³
Melt Density	9.5 lbs/in ³	0.739 g/cm ³
Bulk Density		
Pellets	32 - 36 lbs./ft ³	513 - 577 kg/m ³
Flake	29 - 31 lbs./ft ³	465 - 497 kg/m ³
Permeability Coefficients:		
Water (@25°C)	6.0x10 ⁻⁹ in ² /sec ² -atm	5.1x10 ⁻¹⁰ cm ² /(sec-cm Hg)
Oxygen (@ 30°C)	2.7x10 ⁻⁹ in ² /sec ² -atm	2.3x10 ⁻¹⁰ cm ² /(sec-cm Hg)
Carbon Dioxide (@30°C)	10.8x10 ⁻⁹ in ² /sec ² -atm	3.5x10 ⁻¹⁰ cm ² /(sec-cm Hg)
Hydrogen (@ 20°C)	48.3x10 ⁻⁹ in ² /sec ² -atm	41.0x10 ⁻¹⁰ cm ² /(sec-cm Hg)
Nitrogen (@30°C)	0.52x10 ⁻⁹ in ² /sec ² -atm	0.27x10 ⁻¹⁰ cm ² /(sec-cm Hg)
Water Absorption @24 h Immersion	<0.01%	<0.01%
Brittleness Temperature		
Homopolymer	13 - 32 °F	-10 - 0 °C
Random Copolymer	-5 - 20 °F	-20 - -5 °C
Impact Copolymer	-44 - 20 °F	-40 - -5 °C
Mechanical Properties		
Modulus of Elasticity (Young's Modulus)		
Homopolymer	183,000 psi	1,300 MPa
Copolymer	155,000 psi	1,100 MPa
Poisson's Ratio	0.42	0.42
Hardness		
Shore D Scale	55 - 65	55 - 65
Rockwell R Scale		
Homopolymers	80 - 110	80 - 110
Random and Impact Copolymers	45 - 95	45 - 95
Coefficient of Friction, Plastics to Steel		
Static	0.30	0.30
Dynamic	0.28	0.28
Coefficient of Friction, Plastics to Plastic		
Static	0.76	0.76
Dynamic	0.44	0.44
Thermal Properties		
DSC Melting Point		
Homopolymer	320 - 329 °F	160 - 165 °C
Copolymer	275 - 318 °F	135 - 159 °C
Specific Heat (@ 23°C)		70 - 80 J/°K/mol
Heat of Fusion	37.8 Btu/lb	88 kJ/kg
Thermal Conductivity (solid)		0.17 - 0.22 W/m °K

Press Fit Analysis

Basic Params for Calculations

```
SF = 10; % safety factor [-]
in_to_m_conversion = 39.37;
E = 1300 * 10^6; % young's modulus [Pa]
nu = 0.42; % poisson ratio [-]
```

Energy Calcs for Contact Force Simulation in Fusion

```
rough_mass = .03; % kg
g = 9.81; % m/s^2
h = 47.5 / in_to_m_conversion; % inches above floor for drop test simulation

init_vel = sqrt(2*g*h); % (m/s) initial velocity for fusion simulation

contact_force = 108.8; % N of force at sphere-floor interface
mu_s_p2p = .76; % Static plastic-to-plastic coefficient of friction for PP
```

Shigley's Interference Calculations {02/27/25}

```
R = 1.25 / in_to_m_conversion; % nominal radius [m]
r_o = 1.5 / in_to_m_conversion; % outer radius [m]
r_i = 1.24 / in_to_m_conversion; % inner radius [m]
length_of_engagement = 0.2 / in_to_m_conversion;
```

```
zeta_o = r_o/R;
zeta_i = r_i/R;

C_o = (zeta_o^2 + 1)/(zeta_o^2 - 1);
C_i = (zeta_i^2 + 1)/(zeta_i^2 - 1);
```

```
K_o = (1/E) * (C_o + nu); % constant for yo-yo ring
K_i = (1/E) * (C_i + nu); % constant for yo-yo body
```

```
A_wall = 2*pi*R*length_of_engagement;
```

```
p_sim = contact_force/(A_wall * mu_s_p2p); % contact pressure at wall
p_yield = 33*10^6; % contact pressure at wall
```

```
interference_yield = SF * ((K_o + K_i) * p_yield * R)/in_to_m_conversion
```

```
interference_yield =  
-0.0242
```

```
interference_sim = SF * ((K_o + K_i) * p_sim * R)/in_to_m_conversion
```

```
interference_sim =  
-1.0351e-04
```

Name	Unit	Expression	Value	Comments
wall_thickness	in	0.08 in	0.08	uniform wall thickness
outer_radius	in	1.25 in	1.25	outer radius of kirby
inner_radius	in	outer_radius - wall_thickness	1.17	inner radius of kirby
length_of_engagement	in	0.2 in	0.2	press fit engagement
total_outer_radius	in	outer_radius + floatie_radius	1.75	total size of floatie
nut_spacer_width	in	0.25 in	0.25	space for nut spacer
nut_spacer_height	in	(1 / 8) * 1 in	0.125	space for the nut spacer
floatie_radius	in	0.50 in	0.5	radius of just the floatie
interference	in	0.0242 in	0.024	interference for press fit
foot_heel_radius	in	0.45 in	0.45	radius of the circle composing kirby's foot
foot_length	in	1.65 in	1.65	length of kirby's foot from heel to toe
shrinkage_scaling		(1.0152 in) / in	1.0152	PP shrinkage
draft_angle	deg	2 deg	2	draft angle
floatie_fillte_radius	in	0.13 in	0.13	top edge fillet

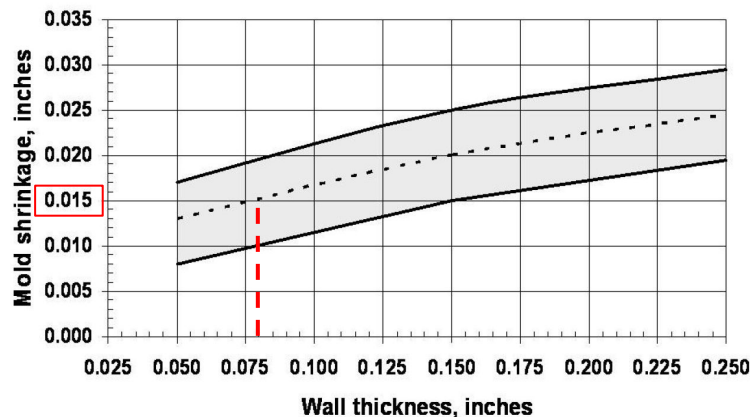


interference	in	0.0242 in	0.024 interference for press fit
--------------	----	-----------	----------------------------------



Shrinkage Analysis

Figure 1: Shrinkage of polypropylene



$$\text{scaling factor} = \frac{1}{1 - \% \text{ shrinkage}}$$

shrinkage_scaling

(1.0152 in) / in

1.0152 PP shrinkage

All parts use shrinkage_scaling in CAD

Name	Unit	Expression	Value	Comments
wall_thickness	in	0.08 in	0.08	uniform wall thickness
outer_radius	in	1.25 in	1.25	outer radius of kirby
inner_radius	in	outer_radius - wall_thickness	1.17	inner radius of kirby
length_of_engagement	in	0.2 in	0.2	press fit engagement
total_outer_radius	in	outer_radius + floatie_radius	1.75	total size of floatie
nut_spacer_width	in	0.25 in	0.25	space for nut spacer
nut_spacer_height	in	(1 / 8) * 1 in	0.125	space for the nut spacer
floatie_radius	in	0.50 in	0.5	radius of just the floatie
interference	in	0.0242 in	0.024	interference for press fit
foot_heel_radius	in	0.45 in	0.45	radius of the circle composing kirby's foot
foot_length	in	1.65 in	1.65	length of kirby's foot from heel to toe
shrinkage_scaling		(1.0152 in) / in	1.0152	PP shrinkage
draft_angle	deg	2 deg	2	draft angle
floatie_fillte_radius	in	0.13 in	0.13	top edge fillet

Criteria

Organization and Planning - delete slide after done

- | |
|--|
| <ul style="list-style-type: none">• please present a preliminary plan for your team to follow that will use the remaining lab sessions to complete the yo-yo project<ul style="list-style-type: none">a. include in this plan a protocol for scaling back high risk features if the team is presented with unmeetable time constraints (i.e. if you are running out of time, struggling, or find out a particular feature can't be made with the resources available, what can you do to simplify your design and still have a yo-yo that meets all of the class requirements?)b. include a list of any problems that you would like assistance from the course staff with! |
| <ul style="list-style-type: none">• students will be graded on<ul style="list-style-type: none">a. the completeness and thoroughness of the required plansb. overall clear messaging, effective presentation, and coherent reasoning presented throughout the Design Reviews |

Manufacturing Timeline



MARCH 2025

SUN	MON	TUES	WED	THURS	FRI	SAT
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17 	18! 	19 	20 	21 	22
23 	24 	25 	26 	27 	28 	29 
30 	31					

Team Goals:



Have all body and ring molds machined for injection molding (3/21)

Meetings



Lab (T 2-5PM)



Team Meeting (7-9PM)



Machining (@ LMP)

Deliverables



Design Review

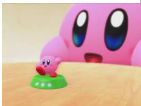
Holidays



Spring Break



APRIL 2025

SUN	MON	TUES	WED	THURS	FRI	SAT
		1! 	2	3 	4	5
6	7	8 	9	10 	11	12
13	14	15! 	16	17 	18	19
20	21 	22! 	23	24 	25 	26
27	28	29 	30			

Team Goals:



- Have all IM Parts Fabricated (4/10)
- Finalize all ring and body molds/solve press fit (4/17)
- Assemble one complete Yo-Yo (4/25)

Meetings



Lab (T 2-5PM)



Team Meeting (7-9PM)

Deliverables



Ring & Body Molds



IM Parts (Insert Molded Nuts)



Manufacturing Review

Holidays



Patriot's Day



MAY 2025

SUN	MON	TUES	WED	THURS	FRI	SAT
				1	2 	3
4	5	6! 	7 	8 	9	10
11	12! 	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Team Goals:



- Assemble 25 Yo-Yos (5/1)
- Finish research poster (5/7)
- Finalize final presentation poster and materials (5/8)

Meetings



Team Meeting
(7-9PM)

Deliverables



50 Yo-Yos (EOL)

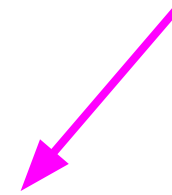


Final Presentation
(12:30-2:00pm)



Manufacturing Contingency Plans

- Team members have tap access/membership at the following makerspaces:
 - MIT Electronics Research Society
 - Morningside Academy for Design
 - Area 51 Shop (CNC Access)
 - SHED
 - Metropolis/The Deep (CNC Access)
 - Pappalardo Lab
- **These spaces will allow us to machine molds outside of LMP hours and do post-machining/3D Printing as necessary.**
- **No outside access for Injection Molding/Thermoforming**



Area 51 Shop



SAFETY
HEALTH
ENVIRONMENTAL
DISCOVERY LAB

Kir-B

Problems for Staff

- Opinion on colors on Kirby Parts during injection molding
- Properly setting injection molding parameters
- Thermoforming Kirby Interior
- General advice on proper CAM changes as the design of molds becomes revised

